Confidence-Guided Joint Complementary Learning for Weakly Annotated Remote Sensing Object Segmentation

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***Abstract*— Object segmentation from weakly annotated remote sensing images (RSIs) is an essential task that helps substantially reduce pixelwise labeling costs. Although mainstream multistage methods have achieved great performance, they generally suffer from high implementation complexity, which limits their practical application. On the other hand, for a single-stage scheme that can be trained in one cycle, error accumulation during joint optimization results in performance degradation. To address these issues, we propose a confidence-guided joint complementary learning (CGJCL) framework for remote sensing object segmentation under image-level annotations that can be easily trained in a single stage and achieve satisfactory performance. CGJCL integrates the localization and segmentation network into a unified framework that employs a joint complementary learning strategy, collaboratively enhancing the performance of each subnetwork. First, a confidence-guided pseudolabel refinement (CGPLR) module is developed to fuse the complementary semantic information of the localization clues and the object boundary/structure details learned from each subnetwork to generate high-quality pseudolabels (PLs) and alleviate error accumulation. Second, a dual self-supervised multiscale consistency (DSMC) loss is presented to both explicitly and implicitly utilize consistency regularization, enabling the promotion of model robustness on multiscale objects and preventing overfitting on false-annotated pixels in noisy PLs. Third, a locally enhanced feature aggregation network is proposed to integrate the multilevel semantic features and alleviate low-level noise interference, producing precise segmentation masks. Extensive evaluation of three RSI datasets demonstrates that the proposed method yields superior performance compared with recent single-stage techniques and several multistage methods, thus revealing its effectiveness and superiority.**

***Index Terms*— Joint complementary learning, object segmentation, remote sensing, weak annotation.**

I. Introduction

OBJECT segmentation, a task aimed at allocating every pixel in the foreground region of an image to the corresponding category, plays a significant role in remote sensing image (RSI) processing [1], [2], [3]. In the past decade, deep learning-based object segmentation approaches [4], [5], [6], [7] have achieved significant advances due to the excellent representation learning ability of deep convolutional neural networks. Nevertheless, the superior performance of the deep learning approach depends considerably on the availability of extensive training samples annotated with pixelwise labels. Moreover, in RSI scenarios with intricate geospatial details and cluttered background interferences [8], [9], [10], precise pixelwise annotations require additional expert knowledge, dramatically increasing the labeling cost and hence limiting the applicability of such approaches.

Recently, object segmentation with weak annotations [11], [12], [13], such as scribbles [14], [15], points [16], [17], bounding boxes [18], [19], and image-level labels [20], [21], [22], [23], which are more convenient to acquire, has become an attractive research topic to alleviate the costly labeling burden of per-pixel labeling [24], [25], [26]. In this article, we use image-level labels, which contain category information that is the easiest to gather, as weak supervision. Naturally, the decreasing labeling cost of weakly annotated segmentation methods will lead to increasing model complexity and more sophisticated training procedures to narrow the performance gap with fully annotated counterparts. More specifically, the prevalent paradigm is to produce coarse object localizations as initial pseudolabels (PLs) from a classification model and then to utilize these PLs to train a fully supervised segmentation network for deriving the final segmentation results [27].

Despite the remarkable performance gains obtained from the stagewise training procedure, there are two main problems with these multistage methods. First, the sophisticated training process must search numerous hyperparameters and preserve the intermediate results of each stage, which consumes immense amounts of space and time and is difficult to implement, thus restricting the practicality of these algorithms [28]. Second, these approaches treat object localization and segmentation as two separate subtasks and individually and sequentially optimize their parameters, which causes the segmentation performance to be heavily dependent on that of the relevant localization network. In addition, stagewise schemes disregard the internal complementary correlation between these tasks and thus limit the potential performance of each task, which can result in suboptimal performance of the whole framework.

To address the issue of high training complexity, early single-stage weakly annotated segmentation methods [29],

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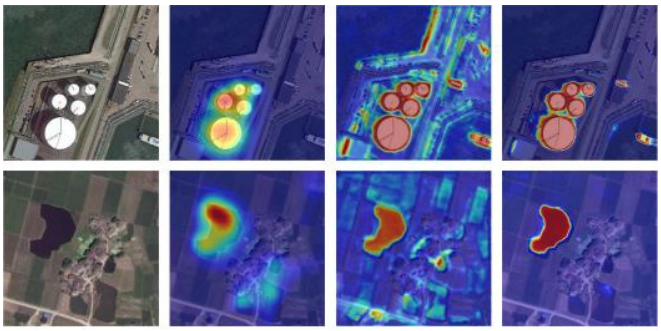

Fig. 1. Illustration of the motivation of our proposed method. (a) Input RSI samples, (b) localization maps obtained via the localization model, (c) segmentation prediction maps generated from the segmentation network by directly employing (b) pseudo supervision, and (d) segmentation prediction results of our CGPLR, which integrates the localization and segmentation models by utilizing their complementary correlation, thus achieving more integral coverage of object areas with more accurate capture of object boundaries.

[30], [31] have been developed, which adopt algorithms such as expectation-maximization, multiinstance learning, and graph-based optimization to infer pixelwise segmentation masks from image labels in a single training cycle. Despite its simplicity, employing a single-stage training scheme to yield satisfactory pixelwise predictions under image-level weak supervision is challenging, resulting in a considerable performance gap between single-stage methods and multistage methods.

Recently, to further increase the performance of single-stage methods, several works have attempted to unify the object localization and segmentation subtasks of multistage schemes into an integral framework in a parallel way [32], [33], [34] since they both require generating pixelwise masks and are natively interrelated. In theory, two networks with diverse architectures and different types of supervision signals will naturally produce distinct and complementary views of feature representation, which can be used to mutually enhance the representation ability of features learned from each task and thus further promote their performance [35]. Specifically, as illustrated in Fig. 1, the object localization network provides clues about coarse object positions, whereas the segmentation network extracts complementary clues, including fine object shapes and structural details with distinct boundaries. To build a connection between the two subnetworks, existing methods [32], [33] utilize the output of the localization subnetwork as pseudo supervision for the segmentation subnetwork.

However, how to effectively integrate the two complementary tasks into a unified end-to-end framework and guide them to learn from each other remains unexplored and challenging in the case of complicated RSIs under weak supervision conditions. First, for weakly annotated RSIs characterized by cluttered interference and blurred object outlines, directly employing the coarse object localization clues output from the localization subnetwork as pseudo supervisions to train the segmentation subnetwork can easily cause error accumulation in the joint learning process and degrade the performance of each subtask, thus reducing the quality of the final segmentation results, especially for object boundary pixels and small target objects. In addition, the output from the segmentation subnetwork containing crucial geographic structure clues, which can feed the localization subnetwork with complementary information about the geographic object spatial structure and boundary information, is underutilized for deriving high-quality PLs in RSI scenes to fully encourage the two subnetworks to learn from each other. Third, due to the lack of precise pixelwise annotations, noise inevitably exists in the generated PLs, which is disregarded in most existing weakly supervised joint learning methods and may significantly degrade the overall model performance.

To address these issues, we propose a novel confidence-guided joint complementary learning (CGJCL) framework for end-to-end object segmentation from weakly annotated RSIs. CGJCL is composed of a shared backbone network as a deep feature extractor and two parallel subnetworks, D_{Loc} and D_{Seg}, for object localization and segmentation, respectively. Considering the unique characteristics of RSIs, a joint complementary training strategy that consists of confidence-guided PL refinement (CGPLR) and dual self-supervised multiscale consistency (DSMC) regularization is proposed to enable the two complementary subnetworks to fully excavate the unified feature representation while collaborating effectively with each other. More specifically, the location and geographic structure knowledge learned from the two subnetworks are adaptively integrated by the CGPLR module using confidence estimates to guide the subnetworks to progressively learn from each other and improve the PL quality to relieve error accumulation. Then, treating the refined PLs as the pseudo supervision for D_{Seg}, the DSMC loss is designed to enforce the consistency between multiscale pixel-level predictions from the two subnetworks while ameliorating the negative effect caused by the unreliable annotated samples in noisy PLs. Therefore, the two subnetworks are mutually enhanced by the proposed joint complementary learning procedure, thus efficiently achieving satisfactory segmentation performance.

The main contributions of this article can be summarized as follows.

1) We develop a novel CGJCL framework for object segmentation from weakly annotated RSIs. By utilizing the potential complementary correlation between the localization and segmentation subnetworks, CGJCL can enable accurate geographic object segmentation with well-preserved boundaries in a single training stage, which strikes a good balance between implementation complexity and model performance.
2) To alleviate error accumulation during joint optimization, a CGPLR module is designed to integrate the object location clues and sharp object outline details learned from the two complementary subnetworks to efficiently generate high-quality PLs of the RSIs, which are guided by confidence scores to extract reliable feature representations and help the two learners benefit from each other.
3) A novel DSMC loss is devised to calculate both explicit and implicit consistency regularization on predictions of multiscale transformed inputs, which helps supplement additional self-supervision to reduce overfitting on

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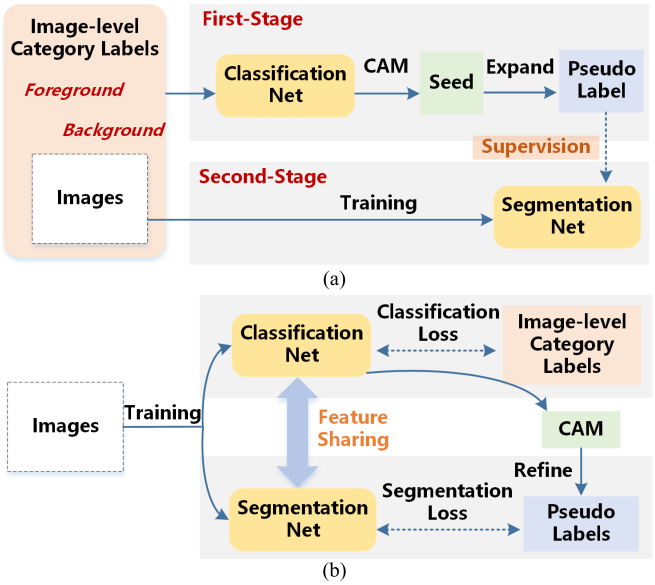

Fig. 2. Mainstream pipelines for (a) multistage segmentation under image-level weak annotations and (b) single-stage segmentation based on joint learning under image-level weak annotations.

false-annotated samples in refined PLs and increases model robustness in detecting geographic objects with various scales.

4) A locally enhanced feature aggregation network is presented to produce precise pixelwise segmentation masks with well-preserved object structures, which effectively utilize multilevel geospatial semantic features and eliminate the noise interference introduced by low-level encoder features.

II. Related Work

In this section, we provide a brief overview of multistage and single-stage object segmentation methods with weak annotations.

A. Multistage Methods

In recent decades, object segmentation with weak annotations, which can achieve pixel-level segmentation predictions by learning from weak supervision, has attracted considerable attention. The majority of commonly employed algorithms leverage multistage training schemes, as shown in Fig. 2(a), which infer class activation maps (CAMs) [21] from a trained classification network with image-level labels and then refine the initial CAMs via various postprocessing operations as PLs to train a subsequent segmentation network to produce accurate pixel-level segmentation results.

However, CAMs suffer from inferior supervision, which lacks specific spatial and shape details of objects, and they can activate only partial regions that play an important role in image classification instead of the integral target regions. Therefore, numerous works have explored more integral and accurate PLs by employing numerous "bells and whistles" to improve segmentation performance. For example, erasing regions with high responses and iteratively learning the remaining regions is a potential solution [36], [37], [38]. Wei et al. [36] proposed a prohibitive segmentation learning method that uses the adversarial erasing mechanism to address the sparse activation issue. Attention mechanisms and combined detection results from different stages or regions are also employed to expand the pseudomask [39], [40], [41], [42]. Wang et al. [39] proposed consistency regularizations on inputs of multiscale transformation to extract self-supervision information to improve network performance. Sun et al. [40] designed a coattention classification network to learn semantic information across images, which enables the network to achieve more comprehensive learning. In [41], the discovered object regions at different training stages are accumulated via an online attention accumulation strategy, which helps gradually increase the integrity of the detected object regions. Li et al. [42] proposed a progressive patch learning method that splits the feature maps into patches for subsequent network processing, thus enhancing the network's response to local regions and yielding more integral activation maps.

Furthermore, auxiliary information such as saliency cues and affinity has been leveraged to obtain high-quality PLs. The explicit pseudopixel supervision method proposed by Lee et al. [43] introduced saliency loss to capture more accurate object boundaries for pseudomasks. Ahn and Kwak [44] designed an extra affinity network to excavate the semantic affinity relationships between neighboring pixels, which can be employed to refine CAMs via random walking operations to increase object coverage. In addition, some researchers have focused on reducing the influence of noise from PLs through denoising loss or label correction strategies [22], [25], [45].

B. Single-Stage Methods

Although multistage segmentation approaches under weak supervision have shown favorable performance, the corresponding cost is a complicated training procedure that necessitates tuning a large number of network parameters at each stage and considerably increases the training complexity of the whole framework [28], [34], [46]. In contrast, single-stage methods usually only require training an end-to-end framework in one training cycle, which usually reduces model performance compared with multistage methods. The early single-stage method, proposed by Papandreou et al. [29], develops an expectation-maximization algorithm, which includes an alternative framework between inferring the pixel-level segmentation labels and optimizing the network parameters. In [30], the image-level weakly supervised learning task was formulated as a multiinstance learning problem, and an aggregation layer was proposed to generate pixelwise labels from image-level supervision.

Recently, researchers have made efforts to improve the performance of single-stage methods. Among them, using joint learning to train the localization and segmentation models has become a prevalent strategy, as illustrated in Fig. 2(b). For example, the reliable region mining (RRM) model proposed by Zhang et al. [32] consists of one classification branch and one segmentation branch and uses the dense conditional random field (CRF) operation [47] to acquire tiny reliable regions in pseudomasks as supervision for the

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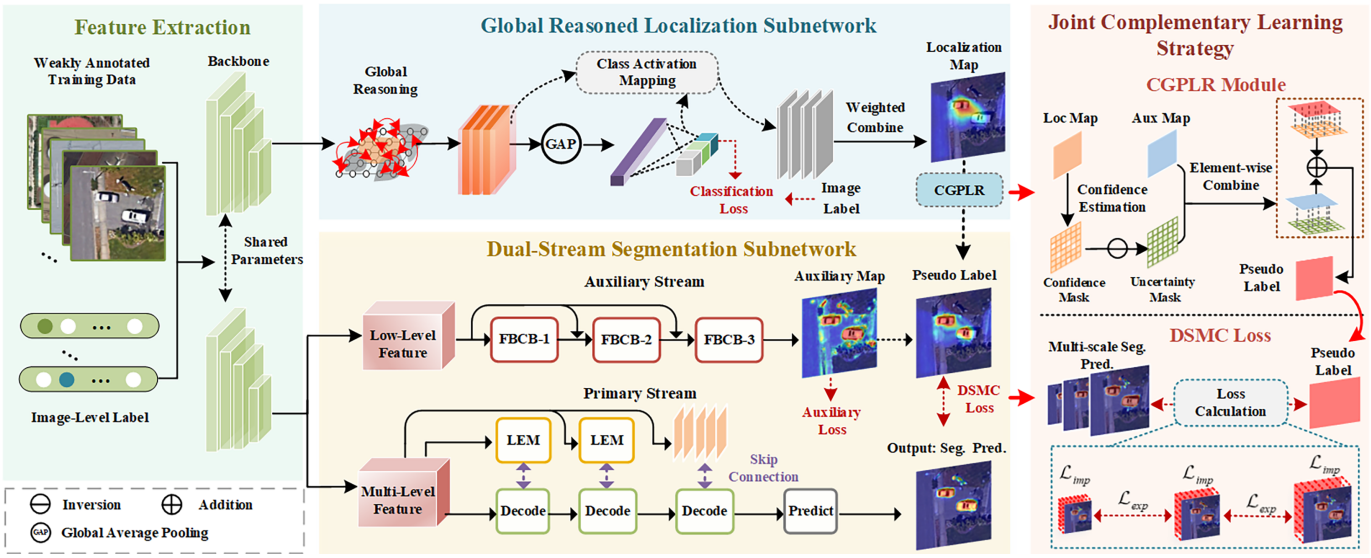

Fig. 3. Flowchart of the proposed method.

segmentation branch. Nevertheless, the CRF-based postprocessing operation is time-consuming and thus substantially increases the computational cost of the algorithm. In contrast, Araslanov and Roth [48] employed a pixel-adaptive mask refinement strategy to efficiently promote the local consistency of the generated mask. Researchers [33] have also introduced a self-paced learning mechanism to the weakly supervised joint learning framework to build stronger deep feature representations for accurate prediction. In [34], a transformer is used as the backbone to learning the feature representation for improving the semantic integrity of the generated PLs, which are used to train the parallel segmentation branch.

Although attention has generally been given to single-stage segmentation under weak annotations for natural scene images, few related studies have been conducted in the field of RSIs. Compared with previous single-stage approaches, our proposed CGJCL considers the characteristics of RSIs in particular to design an effective single-stage training scheme. For example, we propose a CGPLR module that effectively leverages the complementary roles of the two subnetworks for integrating both object locations and sharp object outlines, which are required for the complex RSIs to improve the quality of initial PLs, helping alleviate the error accumulation issue and encouraging the two subnetworks to collaborate effectively with each other. In addition, a locally enhanced feature aggregation network is designed to produce precise segmentation masks, which integrate the multilevel semantic feature representation via skip connections and incorporate local enhancement modules (LEM) to handle the large appearance and scale variations of geographic objects in RSIs. Moreover, a DSMC loss is proposed to calculate the consistency regularization loss for segmentation predictions of multiscale transformed inputs, which increases the model robustness in detecting geographic objects with varying scales and reduces the impact of noisy PLs.

III. Methodology

A. Overview

Let $\{(\mathbf{I}_i, \mathbf{y}_i)\}_i^N$ be a weakly annotated RSI dataset, where N is the total number of training images, $\mathbf{I}_i \in \mathbb{R}^{H\times W\times 3}$ denotes the input RSI, and $\mathbf{y}_i \in \mathbb{R}^{1\times C}$ is the corresponding image-level category label, where C represents the total number of object categories in the dataset. For simplicity, we denote the input image as $\mathbf{I}$ in the following part. As shown in Fig. 3, the proposed CGJCL is composed of four components: a shared backbone network that acts as an encoder for multilevel feature extraction, a localization subnetwork for coarse localization map generation, a dual-stream segmentation subnetwork for precise object segmentation, and a joint complementary learning strategy for training the two complementary subnetworks to achieve mutual performance enhancement.

B. Network Architecture

The proposed CGJCL consists of a shared backbone network followed by two separate network branches for localization and segmentation. The details of the network architecture are described in Section III-B.

1) Backbone Network: The shared backbone for the localization and segmentation subnetworks serves as an encoder to extract hierarchical deep features from the input images. VGG16-Net [49] is a classical convolutional neural network that consists of 13 convolutional layers, five max-pooling layers, three fully connected (FC) layers, and a softmax layer. Here, we modify the structure of VGG16-Net to utilize part of the fully convolutional network as our backbone by removing the last three FC layers and the softmax layer. In addition, to maintain a relatively high resolution of the final output feature activation, we alter the stride of the last two max pooling layers to 1. Thus, the three remaining max-pooling layers with a stride of 2 in the backbone network can natively separate all of the feature maps into four stages by their spatial

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size, denoted as a set of multiscale deep features $\{\mathbf{F}_k^{enc}\}_{k=0}^{3}$ with strides of $\{1, 2, 4, 8\}$, which serve as inputs for subsequent localization and segmentation tasks. The backbone network is fine-tuned with pretrained ImageNet [50] weights.

2) Global-Reasoned Localization Subnetwork: The localization subnetwork D_{Loc} infers the pixelwise object localization maps from the classification model supervised by image-level category labels. Here, we employ the prevalent CAM technique [21] to locate the object regions. In addition, motivated by the strong global contextual representation learning ability of the graph-based global reasoning network, the global reasoning module (GRM) [51] is introduced into the localization subnetwork to enhance the semantic integrity of the generated localization map.

Typically, the GRM is composed of three steps: graph projection, graph reasoning, and graph reprojection. First, the GRM projects an input feature map X from the coordinate space into the interaction space as a feature V representing graph nodes, which is formulated as follows:

$$V = \phi\left(X; W_\phi\right)^T \otimes \theta(X; W_\theta) \tag{1}$$

where both $\phi(\cdot)$ and $\theta(\cdot)$ are 1×1 convolutional layers and $\otimes$ represents the matrix multiplication operation. Then, graph convolution is employed to capture the relationships between features in the interaction space. After reasoning, the output features are projected back into the original space as $\widehat{V}$. The output feature map of the GRM is obtained by using a residual connection between $\widehat{V}$ and the input feature X as follows:

$$X_{out} = W_v \widehat{V} + X \tag{2}$$

where W_v is the parameters of a 1×1 convolutional layer.

Specifically, for our proposed global-reasoned localization subnetwork, the high-level feature $\mathbf{F}_3^{enc}$ serves as the input for the GRM to obtain the global enhanced feature maps. Afterward, the classifier $\mathcal{F}_{\text{CLS}}(\cdot)$ is formed to produce binary classification scores, which consists of a global average pooling (GAP) layer followed by an FC layer. Next, we apply softmax activation to obtain the classification score that denotes the category probability of the input image. The above process is formulated as follows:

$$\mathbf{s} = \mathcal{F}_{\text{CLS}}\left(\text{GRM}\left(\mathbf{F}_3^{enc}\right); \theta_{\text{CLS}}\right) \tag{3}$$

where θ_{CLS} represents the parameters of the classifier.

Finally, the CAM technique is employed to generate the localization map from the localization subnetwork. Specifically, for a given category c, we denote the last convolutional feature map by $\mathbf{F} \in \mathbb{R}^{M \times N \times D}$. The weight matrix of the FC layer in the classifier can be denoted as $\mathbf{W} \in \mathbb{R}^{D \times C}$, where C represents the number of object categories. The CAM $\mathbf{M}_c$ for class c can be obtained as follows:

$$\mathbf{M}_c = \text{ReLU}\left(\sum_{i=1}^{D} \mathbf{W}_{i,c}\mathbf{F}_{:,:,i}\right). \tag{4}$$

The min–max normalization operation is subsequently applied to output the localization map, which is then resized to the same shape as the input RSI sample.

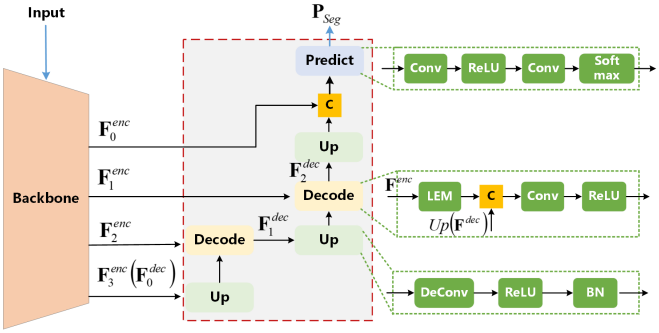

Fig. 4. Architecture of the primary segmentation stream, where "C" denotes the concatenation operation.

3) Dual-Stream Segmentation Subnetwork: To produce pixelwise segmentation predictions, we present a dual-stream segmentation subnetwork that consists of a primary segmentation stream and an auxiliary segmentation stream to fully exploit the extensive feature hierarchies extracted from the backbone to address the large appearance and shape variance of geographic objects in RSIs. The details of the two streams are described below.

a) Primary segmentation stream: The primary segmentation stream aims at restoring the deep encoder features to the size of the input by progressively integrating the shallow and high-level features, thereby predicting the precise pixelwise segmentation masks. In previous single-stage works such as [32], high-level deep features were simply processed by stacked convolutional layers to make segmentation predictions, disregarding the local spatial details that are crucial for detecting geographic objects with complicated background interferences and variable scales. To address this issue, we propose a locally enhanced feature aggregation network as the decoder for primary segmentation prediction. More precisely, we employ skip connections to effectively aggregate the multilevel deep features progressively, thereby compensating for high-level semantic features with the finer details of low-level features. In addition, an LEM is introduced to refine the encoder features before aggregation to eliminate noise interference caused by noisy encoder features.

Formally, as illustrated in Fig. 4, the multilevel deep semantic features $\{\mathbf{F}_k^{enc}\}_{k=0}^{3}$ output from the backbone network are passed to the primary segmentation stream, which serves as a decoder. To construct the decoder architecture, a basic upsampling module $\text{Up}(\cdot)$ is designed to increase the spatial resolution of the feature map by a factor of 2. $\text{Up}(\cdot)$ is composed of a transposed convolution layer (kernel size $= 3 \times 3$, stride $= 2$, and padding $= 1$), followed by a rectified linear unit (ReLU) activation function and a batch normalization (BN) layer.

In the top-down pathway, the decoder features are first upscaled by $\text{Up}(\cdot)$ and then combined with the corresponding locally enhanced encoder features of the same spatial size via skip connections to effectively integrate the coarse semantics with fine structural detail information, which is formulated as

$$\mathbf{F}_1^{dec} = \text{Conv}^*\left(\left[\text{Up}\left(\mathbf{F}_0^{dec}\right), \text{LEM}\left(\mathbf{F}_2^{enc}\right)\right]\right) \tag{5}$$

$$\mathbf{F}_2^{dec} = \text{Conv}^*\left(\left[\text{Up}\left(\mathbf{F}_1^{dec}\right), \text{LEM}\left(\mathbf{F}_1^{enc}\right)\right]\right) \tag{6}$$

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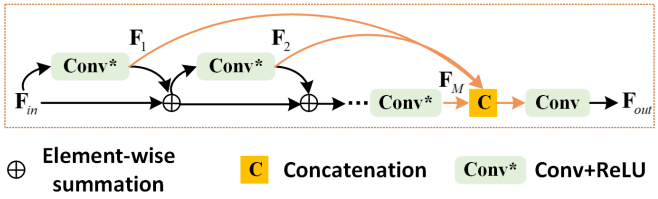

Fig. 5. Illustration of the LEM.

where $[\cdot,\cdot]$ denotes the concatenation operation in the dimension of the channel, and Conv* denotes the convolutional operation followed by an ReLU activation function, $\mathbf{F}_0^{dec} = \mathbf{F}_3^{enc}$. Finally, the last output decoder feature $\mathbf{F}_2^{dec}$ is upscaled and then concatenated with the corresponding encoder feature $\mathbf{F}_0^{enc}$ to estimate the segmentation prediction $\mathbf{P}_{Seg}$ as follows:

$$\mathbf{P}_{Seg} = \text{Softmax}\big(\text{Conv}\big(\text{Conv}^*\big(\big[\mathbf{F}_0^{enc}, \text{Up}\big(\mathbf{F}_2^{dec}\big)\big]\big)\big)\big). \quad (7)$$

We then provide a detailed introduction to the LEM. In a typical encoder–decoder-based segmentation network [52], the decoder and corresponding encoder features are directly integrated via skip connection structures. Nevertheless, for geographic object segmentation, which requires well-preserved spatial details and object outlines, direct combination introduces considerable disturbances from low-level encoder features, which may impair recognition of object boundaries and result in ignorance of tiny targets [53]. To address this issue, the LEM is designed to enhance noisy low-level encoder features and is incorporated into the primary segmentation stream, facilitating the provision of more valuable local details with fewer disruptions that are detrimental to object segmentation.

As shown in Fig. 5, the LEM employs densely connected convolutions to continually transmit low-level local features of preceding blocks to the end of the module, which aggregates the local residual features to generate higher quality feature representations, as inspired by the residual aggregated network [54]. Given the input $\mathbf{F}_{in}$, the corresponding output $\mathbf{F}_{out}$ of the LEM is calculated as follows:

$$\mathbf{F}_i = \begin{cases} \text{Conv}^*(\mathbf{F}_{in}), & i = 1 \\ \text{Conv}^*(\mathbf{F}_{in} + \mathbf{F}_1 + \cdots + \mathbf{F}_{i-1}), & i = 2, \ldots, M \end{cases} \quad (8)$$

$$\mathbf{F}_{out} = \text{Conv}([\mathbf{F}_1, \ldots, \mathbf{F}_M]) \quad (9)$$

where $i = 1, \ldots, M$ represents the index of intermediate blocks in the LEM and M is empirically set to 3 in our implementation.

b) Auxiliary segmentation stream: Although the primary stream is capable of producing pixel-level segmentation masks, the integration of low-level detail features and high-level semantic features incurs certain calculation costs, resulting in unstable and inferior performance during the early training stage. Hence, it is necessary to construct an auxiliary stream with faster convergence to feedback fine structural detail information from the segmentation flow to the PL refinement module described in Section III-C to generate high-quality PLs, with the corresponding experimental validation given in "w/o Aux" of Section IV-C. Here, we construct an auxiliary segmentation stream by fully exploiting the low-level features exhibiting rich spatial details.

Technically, an auxiliary stream consisting of three stacked feedback convolutional blocks (FBCBs) is constructed to rapidly produce a full-resolution segmentation mask. By employing the cross-layer connections in FBCBs proposed by Zhang and Ma [23], the feed-forward propagation speed of the fully supervised auxiliary stream is increased. In addition, in the absence of downscaling operations, the auxiliary stream is capable of preserving the spatial details of the input features.

Specifically, the computation process of the ith FBCB is formulated as (10), in which the cross-layer connected features are processed by three convolutional layers with a kernel size of 3×3, denoted as $f_{\text{FBCB}}^i(\cdot)$, to obtain the output feature of the ith block

$$\mathbf{F}_i = f_{\text{FBCB}}^i\big(\big[\mathbf{F}_{in}, \mathbf{F}_{i-1}\big]\big), \quad i = 1, \ldots, 3 \quad (10)$$

where $\mathbf{F}_0 = \mathbf{F}_{in}$ and the $\mathbf{F}_0^{enc}$ output from the backbone encoder is employed as the input shallow feature $\mathbf{F}_{in}$ for each FBCB. The output feature $\mathbf{F}_3$ of the last FBCB is processed by three stacked convolutional layers followed by a softmax function to yield the auxiliary segmentation predictions $\mathbf{P}_{Seg}^{aux}$, which are formulated as

$$\mathbf{P}_{Seg}^{aux} = \text{Softmax}(\text{Conv}(\text{Conv}(\text{Conv}(\mathbf{F}_3)))) \quad (11)$$

where the last convolutional layer has two channels and a kernel size of 1.

C. CGJCL Strategy

Recently, joint learning mechanisms, such as image classification and object detection tasks for natural scene images [33], [35], have been introduced into the visual recognition community. In a joint learning-based framework, different types of supervision signals are utilized to collaboratively train learners with diverse structures, and consistency-based loss is employed to mutually boost the learning capacity of parallel learners and thus achieve superior overall performance. Nevertheless, compared with natural scenes, RSIs feature more ambiguous object boundaries and cluttered background interference, making it challenging to generate satisfactory predictions with only image-level annotations given. Therefore, in the case of RSI scenes under weak supervision, inaccurate and ambiguous outputs from the localization and segmentation branches easily cause error accumulation and substantially degrade the network performance.

To address the aforementioned issues, a CGJCL strategy is designed to effectively integrate the two subnetworks into a unified framework, fully harnessing the complementary correlation while mitigating error accumulation. First, a confidence-guided PL refinement (CGPLR) module is proposed to effectively aggregate the reliable and complementary knowledge learned from each subnetwork and improve the reliability of PLs for remitting error accumulation. In addition, a DSMC loss is designed to provide additional self-supervision for CGJCL by excavating the internal information from input samples of multiscale image transformation, hence further suppressing the overfitting caused by noisy PLs.

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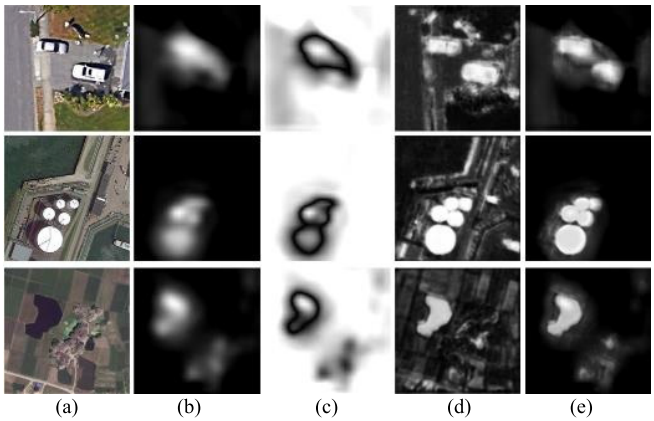

Fig. 6. Visual representations of RSIs. From left to right, the columns represent (a) original RSIs, (b) localization maps, (c) CF of the localization maps, (d) auxiliary segmentation prediction maps, and (e) PLs generated by CGPLR.

1) Confidence-Guided PL Refinement: The classifier-generated localization map generally tends to be oversmooth and suffers from sparseness with indistinct object outlines. Intuitively, training a segmentation network with coarse localization maps as pseudo supervisions appears to yield ambiguous predictions with imprecision. For single-stage work, many approaches [20], [32], [55] utilize CRF operations to eliminate mislabeled pixels in initial PLs, which greatly reduces the training efficiency. Recently, Araslanov and Roth [48] and Ru et al. [34] employed pixel-adaptive convolution operations based on color and spatial information to enhance the local consistency of CAMs, which also requires an iteration process. In contrast, we design an efficient way of refining PLs by utilizing the segmentation output to complement semantic information for localization maps without needing to optimize additional hyperparameters.

As shown in Fig. 6, the auxiliary segmentation predictions accurately exhibit the object outlines while containing many cluttered background fragments caused by cross-layer connections with shallow-layer features. In contrast, the localization maps contain accurate object location clues with blurred boundaries. Based on the above observations, our proposed CGPLR module fully utilizes the complementary output of two subnetworks to generate high-quality pseudo supervision by employing auxiliary segmentation masks with sharp object boundaries to refine the localization maps. In this way, the initial PLs can be refined efficiently, without the requirement of postprocessing operations or optimizing additional hyperparameters. The details of the CGPLR module are provided below.

First, we estimate the confidence values as adaptive weights for integrating the complementary maps. Intuitively, for a CAM-based localization map, pixels with relatively large or small gray values generally tend to be correctly determined, whereas pixels near the segmentation decision boundary are more ambiguous and likely to make incorrect judgments, as reported in [23]. Therefore, the pixels closer to the decision boundary are more unconfident, and vice versa.

In contrast to [23], which uses a fixed threshold, we employ the OTSU algorithm [56] to adaptively calculate the threshold value as the decision boundary to more precisely assign the confidence degree of the prediction probability map. By measuring the distance between the intensity value of a pixel in the localization map and the corresponding segmentation thresholding value of the map, the confidence value of pixel (i, j) can be obtained as follows:

$$\mathbf{CF}(i,j) = \mathrm{Norm}\left(e^{|\mathbf{P}_{Loc}(i,j) - \mathrm{Otsu}(\mathbf{P}_{Loc})|}\right) \tag{12}$$

where Otsu$(\cdot)$ denotes the OTSU thresholding method for obtaining the thresholding value of the input and Norm$(\cdot)$ is the min–max normalization operation.

Afterward, to efficiently aggregate the object location and shape information as a refined PL, we employ the generated confidence map **CF** as a pixelwise guidance map, which increases the sharpness of object outlines for PLs while suppressing vast background interference. As shown in Fig. 7, the refined PL is obtained via a pixel-by-pixel combination of the initial PL $\mathbf{P}_{Loc}$ and the auxiliary branch prediction $\mathbf{P}_{Seg}^{aux}$ as follows:

$$\mathbf{PL}(i,j) = \mathbf{CF}(i,j) \cdot \mathbf{P}_{Loc}(i,j) + (1 - \mathbf{CF}(i,j)) \cdot \mathbf{P}_{Seg}^{aux}(i,j). \tag{13}$$

Considering that uncertainty in deep neural networks typically denotes the model's lack of confidence in its predictions, the inverse operation of **CF** is used for uncertainty value evaluation in (13). In addition, the combination results in (13) are further normalized to the range of [0, 1] via the min–max normalization operation. Through this process, the contributions of the two subnetworks for PL generation can be adaptively adjusted according to the pixelwise confidence estimation values.

To illustrate the rationality of the designs of the proposed CGPLR, heatmaps of the outputs from different streams are shown in Fig. 6. The PLs in Column (e) both accurately locate the foreground regions via the localization clues in Column (b) and preserve the sharp object outlines from the auxiliary maps in Column (d), which fully integrate the complementary knowledge learned from the two learners.

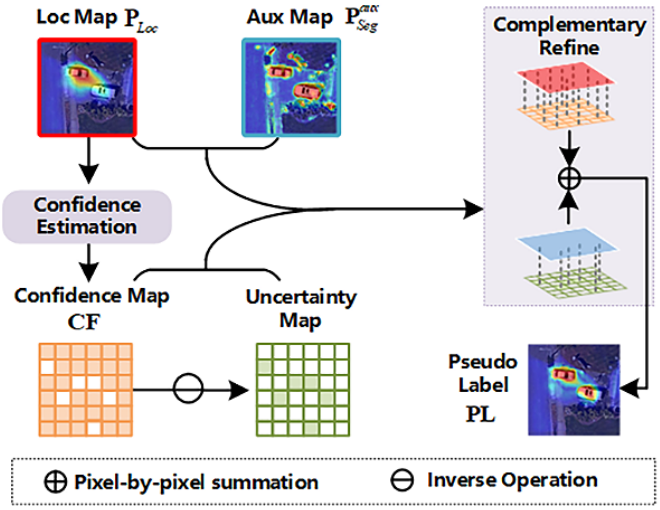

Fig. 7. Flowchart of the CGPLR module.

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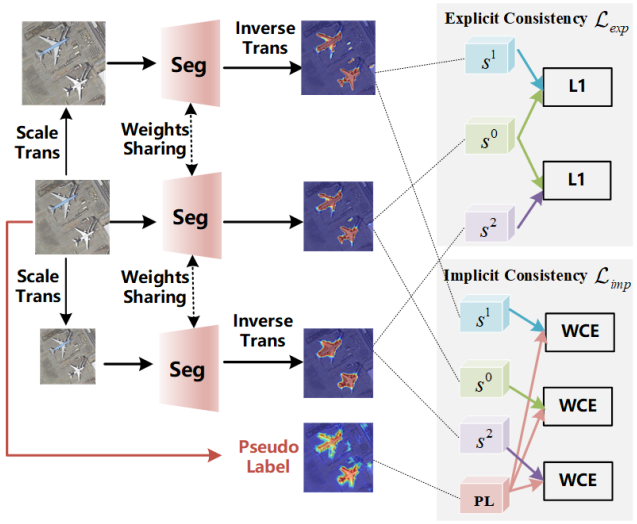

Fig. 8. Flowchart of DSMC loss.

2) DSMC Loss: Although CGPLR has been employed to achieve more reliable pseudo supervision, the noisy label issue persists owing to partial false-corrected pixels resulting from the lack of pixelwise ground truth, particularly for the object boundary regions in complicated RSI scenarios. Hence, introducing a regularization technique to prevent network overfitting on the latent incorrectly refined labels is critical. Consistency regularization [57], [58], a prevalent strategy for semisupervised learning tasks that has recently received much attention from the weakly supervised learning community [39], encourages the model to produce consistent predictions in response to inputs with arbitrary spatial perturbations.

Thus, motivated by the spirit of the consistency regularization mechanism, a novel DSMC loss is incorporated to eliminate the impact of annotation noise, which fully considers both explicit and implicit consistency regularization terms. As demonstrated in Fig. 8, for DSMC loss, training samples under multiscale image transformations are fed to the segmentation network to calculate explicit and implicit consistency loss terms.

First, derived from the classical consistency regularization mechanism, the explicit consistency loss term calculates the distance between the predictions of multiscale transformed inputs and the predictions of the original inputs, which encourages the model to produce consistent predictions in response to inputs with arbitrary spatial perturbations. Formally, we let M be the number of scale transformation operations. In our implementation, we set $M = 2$ (downscaling 0.5 and upscaling 2) and obtain a set of transformed images $\{T_m(\mathbf{I})\}_{m=1}^{M}$ from the original input image $\mathbf{I}$. In addition to predicting the segmentation masks of the original input, the network simultaneously predicts a set of segmentation masks for the transformed images. To calculate the explicit consistency regularization term, first, the predicted masks of the transformed images are transformed back by the corresponding inverse transformation operation $\widehat{T_m}(\cdot)$. The explicit multiscale consistency regularization loss term is then formulated as follows:

$$\mathcal{L}_{\text{exp}} = -\frac{1}{M}\sum_{m=1}^{M} D\left[g(I;\theta), \widehat{T_m}(g(T_m(I);\theta))\right] \tag{14}$$

where $D(\cdot,\cdot)$ denotes the distance measurement operation, which is the $L1$ distance in our experiment, and $g(\mathbf{I};\theta)$ denotes the output of the primary segmentation stream with network parameter θ for the input image $\mathbf{I}$.

Second, as demonstrated in [59], pure explicit consistency is liable to cause a collapse of network learning due to a lack of constraints for each transformed image and to hinder the semantic completeness of the ultimate predictions. Therefore, in contrast to previous consistency regularization-based weakly supervised learning methods such as [39], which merely calculate the explicit consistency regularization between CAMs of various affine transformed images, the proposed DSMC loss further introduces an implicit consistency regularization loss term $\mathcal{L}_{\text{imp}}$ to consider the prediction accuracy of the input in each transformed form and improve the stability of the algorithm.

Specifically, the PL refined by the CGPLR module is employed to provide pixelwise supervision for each transformed prediction, thus enforcing the inherent prediction consistency between multiple transformed scales implicitly as

$$\mathcal{L}_{\text{imp}} = \frac{1}{M}\sum_{m=1}^{M} \mathcal{L}_{\text{WCE}}\left(\widehat{T_m}(g(T_m(\mathbf{I});\theta)), \text{PL}\right) \tag{15}$$

where $\mathcal{L}_{\text{WCE}}$ denotes the weighted cross-entropy loss obtained by utilizing the binary confidence mask of the refined PL as weights, which helps train the network with more confident samples.

Therefore, the total consistency loss term of the primary segmentation stream is expressed as

$$\mathcal{L}_{\text{con}} = \lambda_{\text{exp}} \cdot \mathcal{L}_{\text{exp}} + \mathcal{L}_{\text{imp}} \tag{16}$$

where λ_{exp} represents the weighting coefficient of the implicit consistency loss term, which is set to 4 in our experiment.

Overall, by calculating both the explicit and implicit consistency regularization terms, DSMC loss can encourage the deep network to learn multiscale relationships between objects and to explicitly and implicitly learn the inner semantic information, which provides extra self-supervision for network training with noisy labels.

D. Network Training

To train the localization subnetwork, the multilabel soft margin loss between the image-level category label $\boldsymbol{y}$ and the classification prediction score $\boldsymbol{s}$ is adopted to define the image-level classification loss as follows:

$$\mathcal{L}_{\text{cls}} = -\frac{1}{C}\sum_{c=1}^{C}\left(y_c \log(s_c) + (1-y_c)\log(1-s_c)\right). \tag{17}$$

With regard to the segmentation subnetwork, the training losses include a cross-entropy loss $\mathcal{L}_{aux}$ to train the auxiliary segmentation stream, as given in (18), and a DSMC loss $\mathcal{L}_{\text{con}}$ for the primary segmentation stream, as given in (16)

$$\mathcal{L}_{aux} = \mathcal{L}_{\text{CE}}\left(\mathbf{P}_{Seg}^{aux}, \mathbf{P}_{Loc}\right). \tag{18}$$

The total training loss of our proposed CGJCL is defined as

$$\mathcal{L}_{\text{total}} = \lambda_{\text{cls}}\mathcal{L}_{\text{cls}} + \lambda_{\text{con}}\mathcal{L}_{\text{con}} + \lambda_{aux}\mathcal{L}_{aux} \tag{19}$$

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Algorithm 1 Joint Complementary Learning Optimization Procedure

Input: RSI dataset with image-level labels $\{(\mathbf{I}_i, \mathbf{y}_i)\}_i^N$.
Output: The well-trained CGJCL network with the parameter set Φ.

```
while not convergence do
    Generate the localization map P_Loc from localization subnetwork D_Loc;
    Generate the object segmentation map P_Seg and the auxiliary pre- diction map P_Seg^aux from the dual-stream segmentation subnetwork D_Seg;
    Obtain the refined pseudo label PL by adaptively integrating P_Loc and P_Seg^aux using Eq. (13);
    Jointly update the parameters of D_Loc and D_Seg via the total loss function L_total as given in Eq. (19);
end while
```

where λ_{cls}, λ_{con}, and λ_{aux} denote the weighting coefficient of each loss term; in our implementation, the weights are set to 0.5, 1, and 0.5, respectively.

The entire joint complementary learning optimization process is illustrated in Algorithm 1. During training, all network parameters of the shared layers in the backbone are produced by the total loss, as given in (19), whereas the separate parameters of each subnetwork are updated only with respect to the corresponding separate loss term. In this way, the proposed CGJCL, which consists of two parallel subnetworks that synchronously update their parameters during forward propagation, can be jointly optimized to achieve efficient interactive perception and encourage the two learners to progressively benefit from each other.

IV. Experiments

A. Dataset and Implementation Details

1) Dataset: To validate the effectiveness of our proposed CGJCL, we conduct extensive comparison experiments on three types of RSI datasets, including the ORS dataset, the EORSSD dataset, and the UCMerced-Mini dataset.

1) *ORS Dataset [60]:* This dataset contains 4199 images, where 2000 images are used for training and the others are used for testing. In our experiment, seven types of geographic objects included in the dataset are selected as the target geographic object category to be segmented: airplane, building, lake, playground, ship, storage tank, and vehicle.
2) *EORSSD Dataset [61]:* This dataset includes 2000 images with 1400 images for training and the others for testing. In EORSSD, ten object classes are regarded as the foreground geographic object categories: airplane, building, car, court, golf course, island, river, ship, storage tank, and water.
3) *UCMerced-Mini Dataset:* We select RSIs containing geographic object categories from the UCMerced dataset [62] as our UCMerced-Mini dataset, which includes 1000 images with 800 for training and the others for testing. The object categories are as follows: airplane, baseball diamond, car, golf course, ship, river, building, storage tank, and tennis court.

During the training process, all the input images are resized to $256 \times 256 \times 3$, and we apply horizontal flipping to the training samples as a data augmentation operation to increase the data diversity.

2) Implementation Details: The experiments are performed with PyTorch and executed on a PC with an NVIDIA TITAN Xp GPU. For network training, we use the stochastic gradient descent (SGD) algorithm to optimize the parameters of the localization and segmentation subnetworks, with a momentum of 0.9 and a weight decay of 0.001. The initial learning rate is 0.001, which decreases by 1.25 every 30 epochs during the training process. In addition, our proposed model is trained in an end-to-end manner with a batch size of 4. During the inference phase, only the weights of the segmentation subnetwork are required to obtain the segmentation prediction results. Furthermore, the OTSU algorithm [56] is adopted to convert the probability prediction output from the segmentation subnetwork to the ultimate object segmentation mask. In addition, CRF is applied as a postprocessing operation.

B. Comparison With the State-of-the-Art

We compare our proposed method with eight recently proposed state-of-the-art (SOTA) weakly supervised object segmentation methods, three of which are single-stage models (i.e., activation modulation and recalibration (AMR) [63], affinity from attention (AFA) [34], and self correspondence distillation (SCD) [64]), and the others are multistage models (i.e., self-supervised equivariant attention mechanism (SEAM) [39], class re-activation maps (ReCAM) [65], complementary patch network (CPN) [66], progressively supervised learning (PSL) [23], and one model is enough (OME) [67]). In this experimental section, the methods that accomplish localization and segmentation via a cascaded structure based on the step-by-step strategy are regarded as the multistage type. In contrast, we train the single-stage approaches without the extra processing step of training a separate segmentation network. For a fair comparison, all competing algorithms are trained from scratch on the RSI datasets.

Among the competing methods, AMR rearranges the distribution of feature importance by employing two complementary branches to obtain weighted CAMs, which promotes the excavation of more complete target object regions. AFA and SCD both introduce the transformer architecture as the backbone of end-to-end networks for learning feature representation with higher semantic integrity. They also employ the joint learning training strategy to concurrently optimize the classification and segmentation branches.

In contrast, the competing multistage approaches, including SEAM, ReCAM, CPN, PSL, and OME, rely on stagewise training schemes to progressively improve the quality of supervision signals, which in turn promotes network performance. More specifically, SEAM employs consistency regularization on CAMs to introduce self-supervision information for training the weakly supervised network, which leads to obtaining CAMs with more integrated coverage. ReCAM introduces a softmax CE loss to reactivate the classification model and extract the ReCAM as PLs, which helps reduce the drawbacks of the initial CAMs caused by the use of binary CE loss.

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TABLE I
QUANTITATIVE PERFORMANCE COMPARISON ON THREE RSI DATASETS. "S" DENOTES A SINGLE-STAGE METHOD, AND "M" REPRESENTS A MULTISTAGE METHOD

Methods	Stage	Classification Backbone	Params (M)	Inference Time (s)	ORS			EORSSD			UCMerced-Mini		
					mIoU	PA	mF1	mIoU	PA	mF1	mIoU	PA	mF1
SEAM [39]	M	VGG16	58.06	0.0121	36.41	83.89	49.09	30.01	83.98	45.71	27.47	71.79	44.49
ReCAM [65]	M	ResNet50	89.69	0.0155	31.93	75.08	46.65	29.37	77.52	44.83	29.28	64.57	55.49
CPN [66]	M	ResNet38	105.34	0.0249	22.20	80.32	47.81	21.47	91.06	52.73	22.96	83.89	56.72
PSL [23]	M	VGG16	47.00	0.0167	28.85	89.37	59.16	18.37	94.10	53.32	16.67	75.55	39.39
OME [67]	M	ResNet38	105.34	0.0204	26.18	84.88	52.52	23.38	89.09	57.72	21.65	73.59	52.62
AMR [63]	S	ResNet50	49.56	0.3114	27.25	82.94	38.33	23.53	85.79	36.54	25.71	59.08	49.25
AFA [34]	S	MiT-B1	13.68	0.0834	36.23	89.84	69.33	23.32	88.10	57.62	23.37	79.36	62.49
SCD [64]	S	MiT-B1	13.68	0.0135	36.80	90.89	71.74	25.21	88.36	61.36	25.67	83.31	61.49
Ours (CGJCL)	S	VGG16	17.57	0.2248	36.96	92.73	70.44	27.65	94.76	67.48	30.01	86.64	68.31

CPN uses image pairs with complementary patch regions to excavate more complete foreground activation regions. A curriculum learning-based denoising training loss is designed to reduce the impact of training a segmentation network with noisy PLs in PSL. OME uses image-level label filtering and an uncertainty-driven pseudo mask optimization strategy to achieve multiclass RSI semantic segmentation.

The specific quantitative and qualitative comparison results are as follows.

1) Quantitative Comparison: To conduct quantitative evaluations of our proposed method and other compared SOTA methods, the mean intersection over union (mIoU), pixel accuracy (PA), and mean $F1$ (mF1) are adopted as the objective evaluation metrics.

IoU measures the ratio of the intersection and union between the regions of the ground truth and the model prediction results. By calculating the mean IoU value for all categories in the dataset, we can obtain the mIoU score, which represents the overall segmentation performance for all target categories in the dataset. The mIoU score is defined as

$$\text{mIoU} = \frac{1}{N_c}\sum_{i=1}^{N_c}\frac{\text{TP}}{\text{TP}+\text{FN}+\text{FP}} \tag{20}$$

where N_c represents the total number of object categories in the dataset. In addition, TP, FP, and FN denote the numbers of true-positive, false-positive, and false-negative pixels, respectively, in the segmentation results.

PA evaluates the proportion of the number of pixels with the correct prediction category results to the total number of pixels. It is defined as follows:

$$\text{PA} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}}. \tag{21}$$

mF1 is a comprehensive metric that aims at balancing precision and recall. Precision measures the proportion of the target area in the segmentation image to the correct classification area, whereas recall is the ratio between the target area of the segmented image and the target area of the marked image. The definition of mF1 is given below

$$\text{mF1} = \frac{1}{N_c}\sum_{i=1}^{N_c}\frac{2\times\text{Precision}\times\text{Recall}}{\text{Precision}+\text{Recall}} \tag{22}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{23}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}}. \tag{24}$$

Table I shows the mIoU, PA, and mF1 values of the segmentation results on the three datasets. The quantitative evaluation results in Table I display the superiority of the proposed method in terms of achieving a tradeoff between model performance and training efficiency. Our proposed method achieves the best overall performance, with a 36.96% mIoU score and 92.73% PA on the ORS dataset and a 30.01% mIoU score, 86.64% PA, and 68.31% mF1 on the UCMerced-Mini dataset. On the EORSSD dataset, although the mIoU of our proposal is slightly lower than that of several multistage methods, our method achieves a considerable increase in PA and mF1.

In addition, multistage methods, such as SEAM and ReCAM, generally achieve better performances than do most competing single-stage methods that target natural scene images in terms of comprehensive metrics, which is attributed mainly to their application of alternative stagewise training schemes. Compared with the other multistage methods, PSL performs poorly on the EORSSD and UCMerced-Mini datasets, possibly because it was originally designed for extracting residential areas from RSIs. In contrast, our method, which has a simple end-to-end training scheme, outperforms these multistage methods with substantially more complicated pipelines and outperforms the other single-stage methods, illustrating the tradeoff between model performance and training efficiency. Notably, our network also achieves relatively more stable and robust performance on different types of RSI datasets with geographic objects of varying shapes and sizes.

2) Efficiency Comparison: To evaluate the implementation efficiency of the proposed CGJCL with the competing methods, we calculate the number of parameters required during the whole training stage. For a fair comparison, all of the comparison experiments are carried out in the same environment, and all the input samples are resized to the same size of $256 \times 256 \times 3$.

As shown in Table I, the multistage methods generally reveal higher training complexity than the competing single-stage methods do, which is caused mainly by the sophisticated multistage training scheme, including training a classification model followed by training a separate segmentation model. Single-stage methods, such as AMR, which rely

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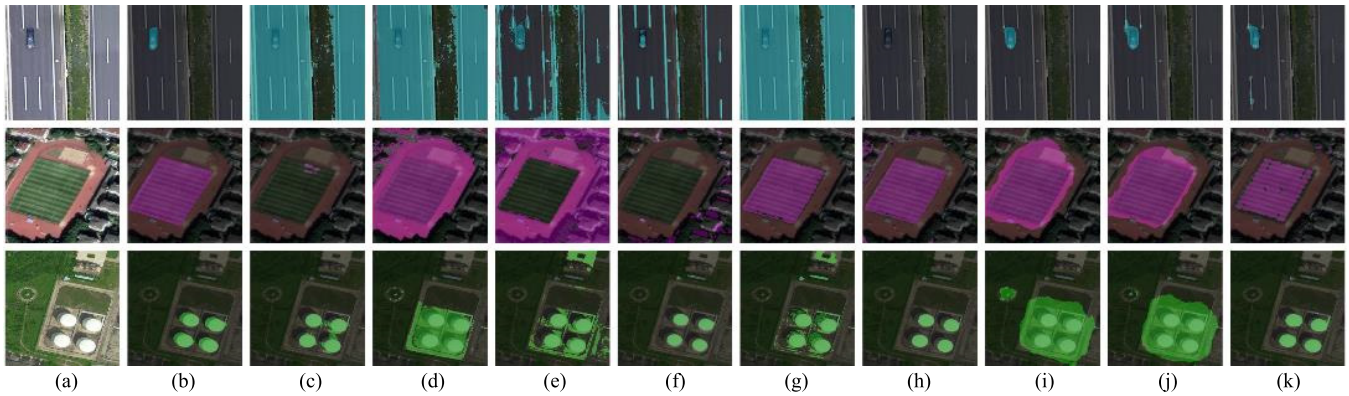

Fig. 9. Comparisons of the object segmentation results on the ORS dataset. (a) Original images. (b) Ground truths. (c) SEAM. (d) ReCAM. (e) CPN. (f) PSL. (g) OME. (h) AMR. (i) AFA. (j) SCD. (k) Ours.

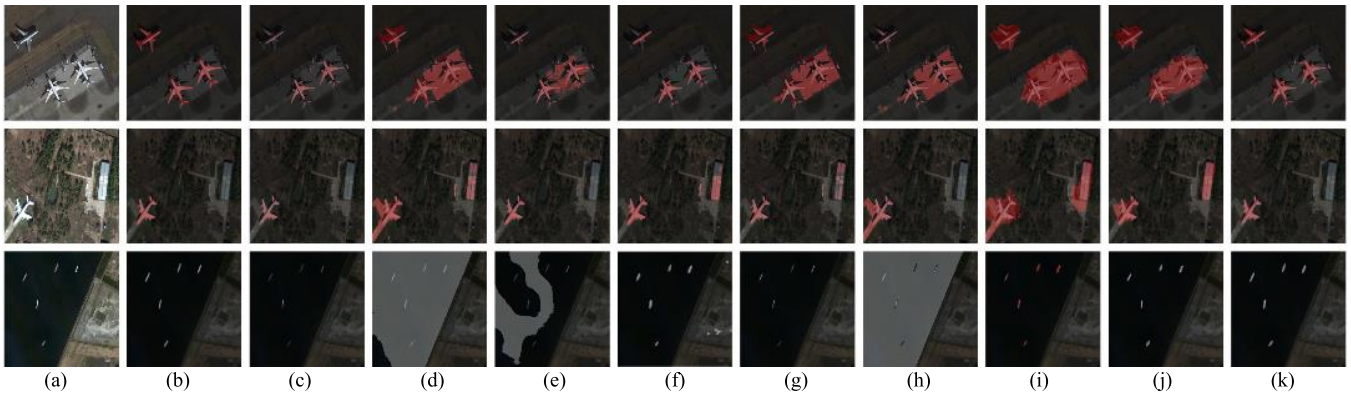

Fig. 10. Comparisons of the object segmentation results on the EORSSD dataset. (a) Original images. (b) Ground truths. (c) SEAM. (d) ReCAM. (e) CPN. (f) PSL. (g) OME. (h) AMR. (i) AFA. (j) SCD. (k) Ours.

on retraining operations on the CAM, also have a high number of training parameters.

Our proposed method requires relatively fewer training parameters than most competing methods do, and it merely requires a single training cycle, which greatly increases its practicality. In addition, although our training efficiency is slightly lower than that of AFA and SCD, our model shows performance improvements on the three RSI datasets.

In addition to the training efficiency, we also compare the running time of each method during the test phase. For single-stage methods, the inference efficiency of our method presents a certain gap with that of AFA and SCD. The main reason for this gap is that they employ a lightweight multilayer perceptron (MLP) as the decoder for segmentation prediction, which benefits from the adopted hierarchical transformer encoder architecture. Notably, although with inferior inference efficiency, CGJCL designs a dual-stream segmentation decoder built upon the CNN architecture to fully exploit the extensive feature hierarchies to more effectively adapt to the RSI scenes and improve the segmentation accuracy for RSIs. Moreover, it outperforms the competing multistage methods in terms of not requiring additional models to be trained and preserving intermediate results (e.g., generated PLs).

Overall, the above discussions demonstrate the tradeoff between implementation complexity and model performance of the proposed CGJCL method.

3) Qualitative Comparison: The object segmentation results for several typical images in the ORS, EORSSD, and UCMerced-Mini datasets are provided in Figs. 9–11, respectively.

SEAM and ReCAM fail to activate the background regions that occur together with the target objects, such as the highway and water areas. In the case of CPN, overprediction can be relatively alleviated, but background fragments still exist in the segmentation results. PSL, which is designed for extracting residential areas, displays inferior performance for multiclass geographic objects with various shapes and sizes. In contrast, OME is aimed at multiclass object segmentation for RSIs, which can locate the target regions. Background interferences with colors similar to those of the target objects in complicated RSI scenes, however, can still be incorrectly recognized as false-positives in the final segmentation results, as shown in the second row of Fig. 10. AMR shows relatively clear boundaries but fails to locate small-scale geographic objects, as shown in the first row of Fig. 9 and the second row of Fig. 11. The transformer-based end-to-end methods, including AFA and SCD, can locate the object regions and suppress background interference, but the contours of the segmentation results remain unmatchable with ground truth masks.

In contrast to the competing methods, our proposed CGJCL method shows superior performance in terms of accurate geographic object localization and precise boundary maintenance; it especially outperforms the competing methods in accurately extracting geographic objects with varieties of scales and shapes, such as relatively small-scale vehicles and

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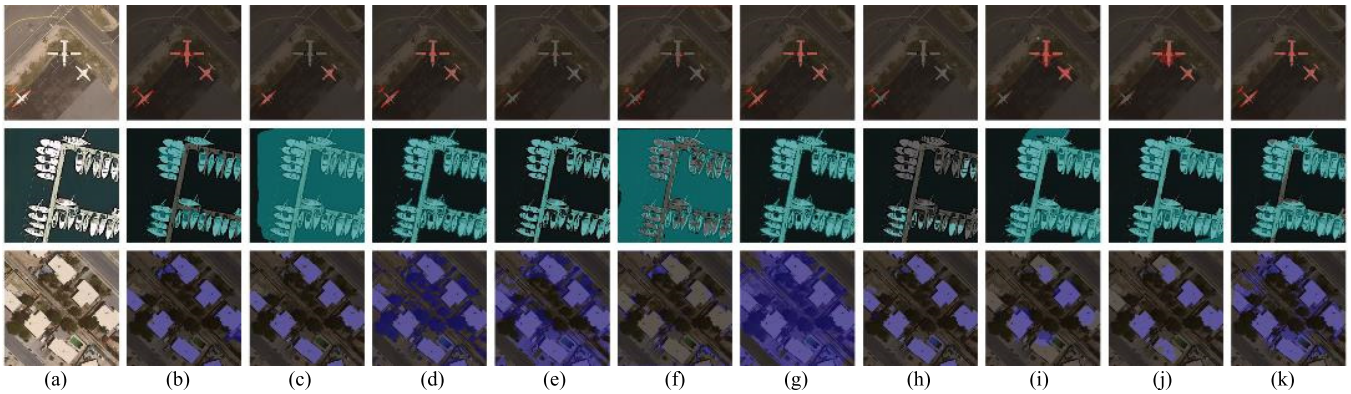

Fig. 11. Comparisons of the object segmentation results on the UCMerced-Mini dataset. (a) Original images. (b) Ground truths. (c) SEAM. (d) ReCAM. (e) CPN. (f) PSL. (g) OME. (h) AMR. (i) AFA. (j) SCD. (k) Ours.

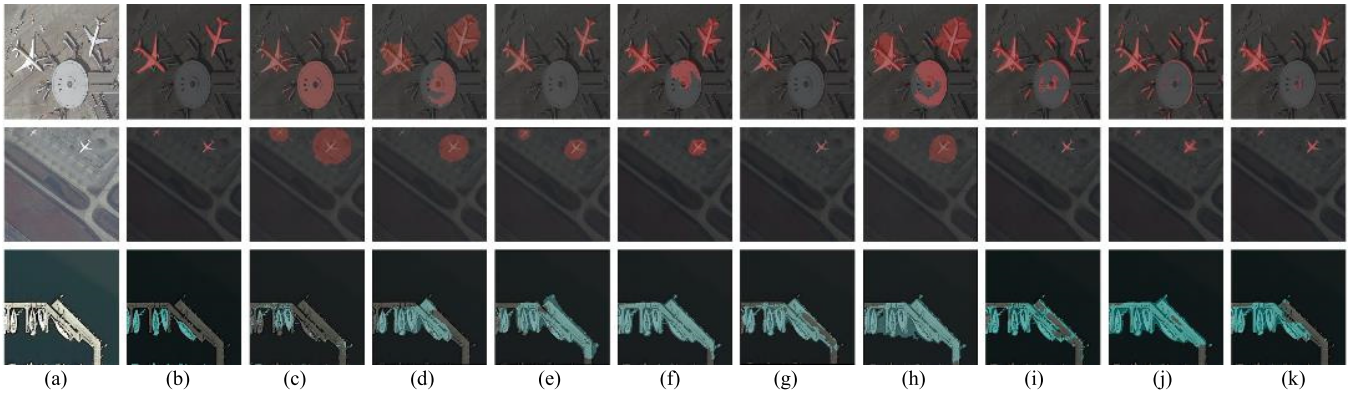

Fig. 12. Object segmentation results of ablation studies on the joint complementary learning strategy and CGPLR module of our proposed model. (a) Original image. (b) Ground truth. (c) 2Stage. (d) w/o CGPLR. (e) w/o Aux. (f) Sum PLR. (g) Mul PLR. (h) w PAR. (i) w/o GRM. (j) w/o LEM. (k) Ours.

ships, as illustrated in the first row of Fig. 9 and the third row of Fig. 10.

In addition, the proposed CGJCL method exhibits excellent performance in distinguishing the target objects from the specific co-occurring background regions, as shown in the first row in Fig. 9 and the second row in Fig. 11. The experimental results demonstrate that the complementary enhancement between the two subnetworks significantly promotes model performance.

C. Ablation Study

This section discusses the comprehensive experiments that were conducted to explore the effectiveness of each component of our proposed framework.

1) Influence of the Joint Complementary Learning Strategy: To validate the effectiveness of our proposed end-to-end weakly supervised object segmentation framework, we compare it with the corresponding multistage counterpart scheme named "2Stage." Specifically, "2Stage" follows the pipeline of mainstream multistage methods to train the weakly supervised localization model and fully supervised segmentation model separately in a cascaded pattern. More specifically, the PLs generated from the well-trained localization network are employed to train a fully supervised segmentation network from scratch.

The quantitative and qualitative results presented in Fig. 12 and Table II indicate that our end-to-end joint complementary learning framework outperforms "2Stage" with considerable increases in the objective indicators. For the ORS dataset, the employment of the joint learning training scheme yields a remarkable increase in the mIoU and mF1 scores; for the EORSSD dataset, the gains are 6.75%, 6.91%, and 15.34% for the mIoU, PA, and mF1 scores, respectively. For the UCMerced-Mini dataset, we obtain increases of 4.06%, 1.57%, and 8.85% in the mIoU, PA, and mF1 scores, respectively. As shown in Fig. 12, compared with "2Stage," our model is capable of eliminating more background interference. The reason may be that mutual enhancement between the two complementary learners helps them learn more powerful and unified feature representations, resulting in a significant improvement in overall performance.

Moreover, the localization maps generated from the localization subnetwork of our CGJCL method are also compared with the CAMs obtained from a separate localization network in Fig. 13. We observe that the joint complementary learning strategy promotes the activation of more integral object regions and obtains more accurate localization maps, indicating the effectiveness of joint complementary learning in enhancing the performance of D_{Loc} via mutual enhancement between the two learners.

The mIoUs of the localization and segmentation subnetworks during the joint complementary learning process are given in Fig. 14. The results show that the performances of the two subnetworks are progressively boosted. Specifically,

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TABLE II
QUANTITATIVE PERFORMANCE COMPARISON OF ABLATION STUDIES ON THREE RSI DATASETS

Dataset	Metrics (%)	2Stage	w/o CGPLR	w/o Aux	Sum PLR	Mul PLR	w PAR	CE Loss	Exp Loss	w/o GRM	w/o LEM	Ours
ORS	*mIoU*	25.41	29.14	34.43	34.44	24.24	28.51	30.72	32.48	35.42	34.98	36.96
	PA	92.30	90.19	93.91	92.94	92.53	90.30	89.63	91.06	91.07	90.64	92.73
	mF1	54.19	59.68	67.31	67.34	51.36	59.45	60.10	62.85	68.72	66.50	70.44
EORSSD	*mIoU*	20.90	20.06	19.95	22.18	21.39	20.52	22.54	25.27	26.83	26.15	27.65
	PA	87.85	91.63	93.31	92.41	97.40	91.34	91.11	93.91	92.40	90.41	94.76
	mF1	52.14	58.15	54.00	58.45	58.66	52.91	57.54	63.95	65.79	63.70	67.48
UCMerced -Mini	*mIoU*	25.95	25.93	25.41	27.33	28.45	26.62	26.07	26.96	28.03	26.92	30.01
	PA	85.07	81.98	80.25	84.52	85.91	80.32	81.06	82.87	83.99	82.71	86.64
	mF1	59.46	61.62	60.92	64.46	66.33	62.86	62.55	63.87	65.36	64.59	68.31

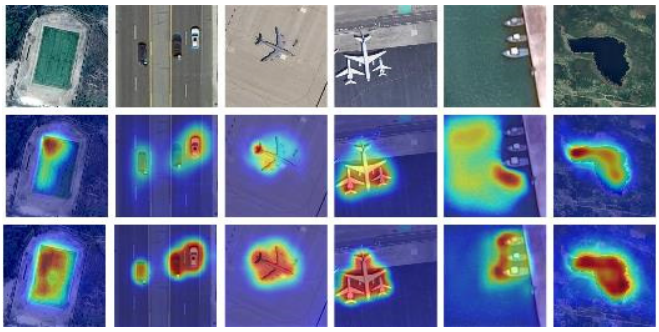

Fig. 13. Comparison of localization maps generated from the separate weakly supervised localization network (second row) and our framework (third row).

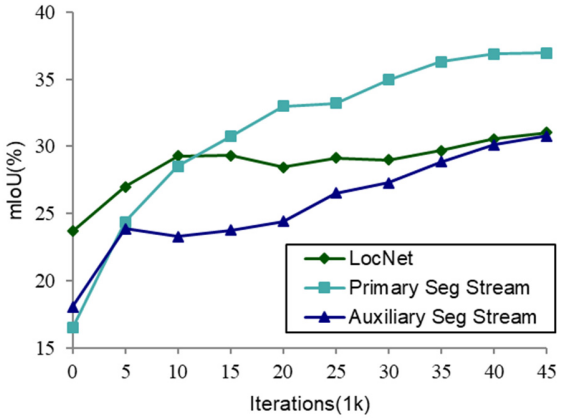

Fig. 14. Changes in the mIoU during the joint complementary learning process.

at the early training stage, D_{Loc} obtains higher IoU scores than both streams of D_{Seg} do, and the auxiliary segmentation stream achieves higher IoU scores than the primary stream does. As the iterations proceed, the performance of the primary segmentation stream rapidly increases and outperforms that of D_{Loc}, which also steadily surpasses the auxiliary stream, demonstrating the effectiveness of the joint complementary learning mechanism for weakly supervised segmentation.

2) Influence of the CGPLR Module: To reveal the effectiveness of the proposed CGPLR module, we design five variants as replacements of CGPLR to generate pseudo supervision for the segmentation subnetwork, including 1) directly utilizing the localization map output from D_{Loc} as PLs (w/o CGPLR); 2) straightly summed averaging or multiplying the localization map and segmentation map as PLs, denoted as "sum PLR" and "mul PLR," respectively; 3) removing the auxiliary segmentation stream from D_{Seg} and integrating the output of the primary segmentation stream with the localization map as PLs to research the validity of the auxiliary stream (w/o Aux); and 4) replacing the CGPLR module by utilizing the PAR module to refine the PL, denoted as "w PAR."

As illustrated in Fig. 12 and Table II, our model outperforms the other five alternatives. Specifically, "w/o CGPLR" can locate mainly the target regions, but it still shows fairly blurred boundaries of the detected objects and exhibits many false-positive results. The reason is that in "w/o CGPLR," the use of coarse localization maps with blurred boundaries as supervision leads to unsatisfactory segmentation output. The performance of "w/o Aux" is still worse than that of our method with CGPLR because the unstable performance of the primary segmentation stream during the early training stage may have a negative impact on the generation of the refined PL. Moreover, both "sum PLR" and "mul PLR" disregard the misalignment between the localization and segmentation branches, and their combination patterns, which apply the same weights to each pixel of the two maps, result in inferior performance. In addition, compared with "w PAR," our proposed CGPLR module exhibits better quantitative and qualitative performance. The above findings display the rationality of the design of the CGPLR module in the case of RSI scenarios by adaptively integrating the localization map and auxiliary segmentation map using confidence estimated scores as weights.

3) Contribution of DSMC Loss: To reveal the contribution of DSMC loss, we replace the proposed DSMC loss with a normal CE loss and a pure explicit consistency loss. In Section IV-C3, we use "CE loss" and "Exp loss" to denote the above two experiments. Note that in addition to the typical explicit consistency term given in (14), the weighted CE loss term between the primary segmentation predictions of the original scale and the PLs was also employed in the "Exp loss" experiment, which is essential for incorporating the two subnetworks as a joint learning framework.

The results demonstrate that by applying both explicit and implicit consistency regularization, our method improves the performance of both "Exp loss" and "CE loss." Fig. 15 shows that the two variants either produce many false-positive detections or show imprecise object outlines. Conversely, our

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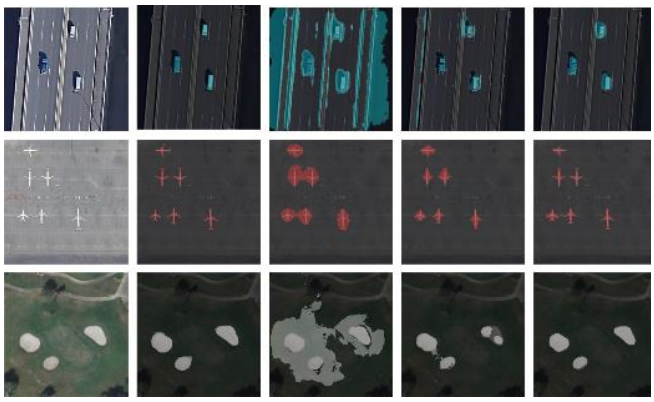

Fig. 15. Comparison of the segmentation results of the variants in terms of DSMC loss. (a) Original image. (b) Ground truth. (c) CE loss. (d) Exp loss. (e) DSMC loss (Ours).

method using DSMC loss is capable of preferably distinguishing geographic objects of varied scales and detecting more accurate contours. More precisely, "CE loss" detects many background interferences since it directly calculates the pixelwise cross entropy between the PL and the segmentation map without filtering out the unreliable pixels and thus introduces noise to the network training. The proposed CGJCL scheme achieves higher objective evaluation results, as shown in Table II, which convincingly indicates its comprehensive superiority to normal CE loss. In addition, the inferior performance of "Exp loss" validates the effectiveness of the introduced implicit consistency term in our DSMC loss.

4) Influence of the GRM: To reveal the contribution of the GRM, we conduct ablation experiments by removing the GRM, denoted as "w/o GRM." Our model outperforms "w/o GRM" in detecting more precise object regions, which confirms the effectiveness of introducing the GRM to the localization network. The main reason is that the ability of the GRM to capture global context relationships increases the semantic integrity of the generated localization maps and hence further increases the segmentation performance.

5) Influence of the LEM: To investigate the effectiveness of our proposed LEM, we conduct an ablation study in which the LEM is removed from the primary segmentation stream of CGJCL, denoted as "w/o LEM." According to the quantitative and qualitative results, there are fewer false-positives and false-negatives in our results than in the "w/o LEM" results, and our proposed method achieves higher comprehensive performance in terms of the quantitative evaluation metrics, which demonstrates the validity of the LEM architecture. The reason may be that the direct combination of encoder and decoder features in "w/o LEM" easily introduces considerable noise to disturb the structural details, resulting in ambiguity between different objects as well as ignorance of small targets.

V. Conclusion

This article proposed a novel end-to-end CGJCL framework to achieve accurate object segmentation from weakly labeled RSIs requiring only a single training stage. CGJCL addresses the implementation complexity of prevailing multi-stage weakly supervised segmentation approaches from a joint complementary learning perspective, which can concurrently achieve object localization and segmentation. With the CGPLR module and DSMC loss, the proposed framework enables the collaboration of the two complementary subnetworks. The CGPLR module employs the estimated confidence maps as weights to adaptively integrate the reliable and complementary knowledge learned from the parallel subnetworks as pseudo supervision so that more unified and powerful feature representations for both tasks can be learned. Then, the DSMC loss leverages both explicit and implicit consistency regularizations to further enforce the perceptual consistency between the two subnetworks and the multiscale segmentation predictions, which supplements comprehensive self-supervision and hence alleviates the impact of noisy PLs. Extensive experiments and an ablation study demonstrate the effectiveness of the proposed method, both quantitatively and qualitatively.

In future research, we aim to investigate methods and design a network architecture to further improve the inference efficiency of the single-stage weakly supervised object segmentation model. In addition, extending the proposed single-stage framework to more areas of computer vision holds significant potential.

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